

Numbers & Objects

UChicago Political Science Math Camp

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A set may consist of numbers, qualitative elements, or anything in between, and is contained by curly brackets.

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A set can even consist of other sets, or of nothing at all.

$$\emptyset$$

Sets in Political Science

- ▶ All countries (by some given definition)
- ▶ All countries recognized by China
- ▶ Legislators on a committee
- ▶ Voters in a society
- ▶ Possible GDP values
- ▶ Possible GDP-per-capita values

Properties of Sets

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- ▶ A set with a cardinality of 1 is called a **singleton**.
- ▶ We can use cardinality to determine **finite** versus **infinite** sets:

$$\#\{1, 2\} = 2$$

$$\#\{\text{All numbers between 1 and 2}\} = \infty$$

The Empty Set

We define the **empty set**, a set containing no elements, as $\emptyset$.

Examples include:

- ▶ The set of all autocracies with free and fair elections
- ▶ Former Soviet states in Africa
- ▶ Senators representing the District of Columbia

Set Operations

We can (and often do) work with **set operations**, which return certain elements in some range of sets.

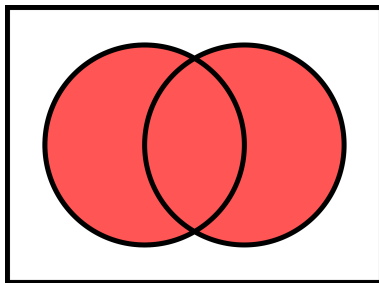
Let

$$A = \{1, 2, 3, 4, 5\} \quad B = \{1, 3, 5, 7, 9\}$$

@Ryan write this on the board

Union

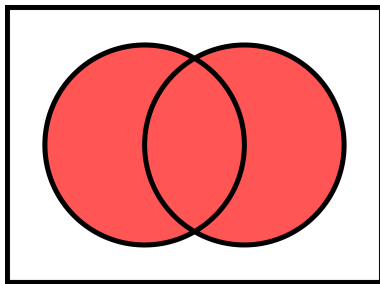
The **union** of A and B is written as $A \cup B$ and defines all elements in *either* A or B .



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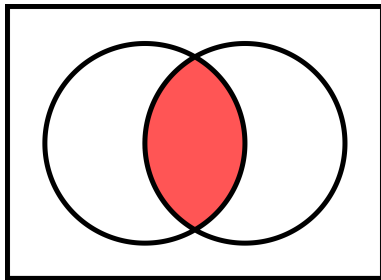


What is $A \cup B$?

$$A \cup B = \{1, 2, 3, 4, 5, 7, 9\}$$

Intersection

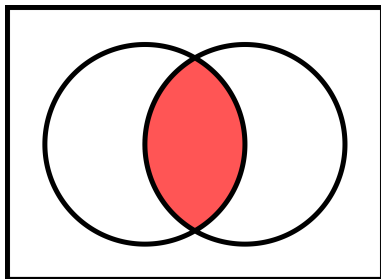
The **Intersection** of A and B is written as $A \cap B$ and defines all elements in *both* A and B .



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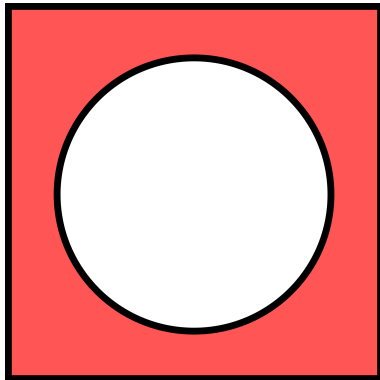
What is $A \cap B$?

$$A \cap B = \{1, 3, 5\}$$

Sample Space & Complement

The **sample space** is all of the possible outcomes being considered.
(Sometimes called the **universe**)

The **complement** of a set A is written as A^C and defines every element of the sample space that is *not* in an element of A .



Complement Examples

Let the sample space be defined as $S = \{1, 2, \dots, 10\}$. Define the following:

► A^C

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- ▶ A^c

- ▶ B^c

- ▶ $(A \cup B)^c$

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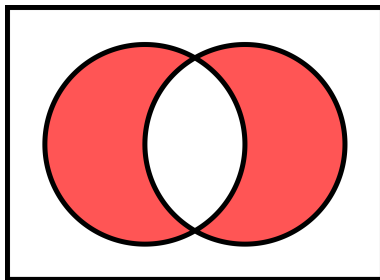
► B^c

► $(A \cup B)^c$

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Symmetric Difference

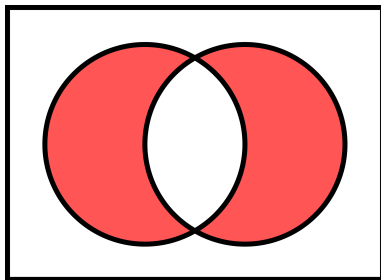
The **Symmetric Difference** of A and B is written as $A\Delta B$ and defines all elements in *either* A and B , but *not* in *both* A and B .



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What is $A\Delta B$?

$$A\Delta B = \{2, 4, 7, 9\}$$

Set Operations

Let us define $S = C$ as the set of counting numbers from 1 to 10, and

$$A = \{1, 2, 3, 4, 5\} \quad B = \{1, 3, 5, 7, 9\}$$

► Union:

$$A \cup B? \quad B \cup C?$$

► Intersection:

$$A \cap B? \quad B \cap C?$$

► What if $D = \{2, 4, 6, 8, 10\}$? $B \cup D?$ $B \cap D?$

Set Operations

- ▶ Excluding:

$$A \setminus B? \quad C \setminus B?$$

- ▶ Complement:

$$A^C? \quad B^C?$$

- ▶ Symmetric Difference:

$$A \Delta B? \quad B \Delta C = ?$$

Back To Elements

We use the symbol $\in$ to represent \in:

$$x \in A \implies x \text{ is an element of } A$$

Similarly, we use $\notin$ to represent \notin:

$$x \notin A \implies x \text{ is not an element of } A$$

Subsets

Some set m is a **subset** of set n if every element in m is also an element in n .

$$A \subseteq C$$

Some set m is a **proper subset** of n if every element in m is also an element in n and there exists some element in n that is not in m .

$$A \subset C$$

This implies:

$$C \subseteq C$$

$$C \not\subset C$$

Reframing the Condition

We can use the definition of a subset to create a logical “test”:

*M is a **subset** of set N if every element in M is also an element in N.*

$\implies$ **If every element in M is also an element in N, then $M \subseteq N$.**

Formally,

$$(x \in M \implies x \in N) \implies M \subseteq N$$

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What about proper subsets?

$$\left((x \in M \implies x \in N) \text{ and } (y \in N \not\Rightarrow y \in M) \right) \implies M \subset N$$

A Preview to Formal Logic

We often want to categorize conditions as **necessary**, **sufficient**, or both/neither:

- ▶ A condition is **necessary** for an outcome if it **MUST** be met in order for the outcome to be true.
- ▶ A condition is **sufficient** for an outcome if its being met **REQUIRES** that the outcome is true.

Question: Which is stronger?

Tying Back

Consider the following condition:

Every element of m is also an element of n

Is this condition necessary and/or sufficient for the following outcomes?

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An important conclusion: $m \subset n$ implies $m \subseteq n$, but $m \subseteq n$ does not imply $m \subset n$. “Proper Subset” is a stronger condition.

Power Sets

The **power set** is the set of all subsets of a set (phew).

Let $n = \{A, B\}$. Then,

$$P^n = \left\{ \{A, B\}, \{A\}, \{B\}, \{\emptyset\} \right\}$$

Let $m = \{A, B, C\}$. Then,

$$P^m = \left\{ \{A, B, C\}, \{A, B\}, \{B, C\}, \{A, C\}, \right. \\ \left. \{A\}, \{B\}, \{C\}, \{\emptyset\} \right\}$$

Note that $\# [P^X] = 2^{\# [X]}$.

Your turn:

$$A = \{4, 5, 6\}$$

$$B = \{5, 6, 7\}.$$

Identify the following and explain what each refers to:

► $A \cup B$

► $A \cap B$

► $B \setminus A$

► $A \Delta B$

► $\#P^B$

Your turn:

$$A = \{51 : 59\}$$

$$B = \{\text{Ellis, University, Woodlawn, Kimbark, Kenwood}\}.$$

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An Illustrating Example

The European Union / Eurozone / Schengen Area

A Step Back: Some Important Notation

We are learning a *language* that is precise, but not necessarily unique.
There are a few key symbols and phrases necessary to understand sets:

- ▶ $\in$ = “in”: $x \in A$ means x is an element in A

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- ▶ (a, b) = all elements from a to b (exclusive)
- ▶ $:$ read as “such that” (sometimes just s.t.)

A Step Back: Set Builder Notation

In order to fully understand the most important sets, we must understand how to write and read in **set-builder notation**.

A common format might look like this:

$$\text{name} = \{x \in \text{parent} : \text{condition 1, condition 2, ...}\}$$

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Let us go back to the zoo example:

$$\text{zoo mammals} = \{x \in \text{zoo animals} : \text{warm-blooded, vertebrate, has hair...}\}$$

The Fundamental Sets

There are four fundamental sets in mathematics: $\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$.

Social scientists should understand $\mathbb{R}$ and $\mathbb{N}$.

Quantitative social scientists should probably think about $\mathbb{Q}$ every once in a while.

$\mathbb{Z}$ is extremely useful in applied mathematics, but doesn't come up much for us.

The Real Numbers

- ▶ Bulky definition, but: all well-defined numbers that can be placed on the number line
- ▶ Expressed as $\mathbb{R}$

The vast majority of political scientists only work with real numbers.

The Natural Numbers

- ▶ The “counting numbers”
- ▶ Expressed as $\mathbb{N}$

Note that the natural numbers are strictly positive!

The Natural Numbers – More Fun

But, why stop there?

$$\mathbb{N} = \{x \in \mathbb{Z} : x + 1 > 1\}$$

$$\mathbb{N} = \{x \in \mathbb{Z} : x \in [0, \infty)\}$$

$$\mathbb{N} = \{x \in \mathbb{Z} : x \in \mathbb{R}, x > 0\}$$

$$\mathbb{N} = \{x \in \mathbb{Z} : x \in \mathbb{R}_{++}\}$$

*There's usually many ways to write a set, and we want to choose the **simplest rigorous option**.*

The Rational Numbers

- ▶ All numbers that can be expressed as a fraction
- ▶ Formally:

$$\mathbb{Q} = \{x = \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$$

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- ▶ Formally:

$$\mathbb{Q} = \{x = \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$$

- ▶ Note that

$$\mathbb{Q} \subset \mathbb{R}$$

And since $1 \in \mathbb{Z}$,

$$\mathbb{N} \subset \mathbb{Q}$$

The Integers

- ▶ The “counting” numbers, both positive and negative
- ▶ Expressed as $\mathbb{Z}$, and defined as:

$$\mathbb{Z} = \{x = a - b : a, b \in \mathbb{N}\}$$

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- ▶ Note that $\mathbb{Z} \subseteq \mathbb{R}$, but NOT $\mathbb{R} \subseteq \mathbb{Z}$. Every natural number is real, but not all real numbers are natural.
 - ▶ Note: this implies that $\mathbb{Z}$ is a proper subset of $\mathbb{R}$!

Above or Below 0

Though we can use

$$A = \{x \in \mathbb{R} : x \geq b\}$$

to limit to only a specific subset, we are often most interested in the case where $b = 0$. Instead, we can express:

$$\mathbb{R}_+, \mathbb{R}_-$$

as shorthand for non-negatives (≥ 0) and non-positives (≤ 0), and

$$\mathbb{R}_{++}, \mathbb{R}_{--}$$

as shorthand for positives (> 0) and negatives (< 0). Note that $\mathbb{N}_- = \emptyset$

Your Turn:

Use set-builder notation to express each of the following in two different ways, making note of the simplest rigorous option:

- ▶ Example: All even numbers

Your Turn:

Use set-builder notation to express each of the following in two different ways, making note of the simplest rigorous option:

- ▶ Example: All even numbers
- ▶ Every number between e and π
- ▶ All counting numbers less than 81
- ▶ All integers that are the square root of another integer
- ▶ All real, irrational numbers

Defining Infinity

- ▶ Each of the four primary sets $\mathbb{R}, \mathbb{Z}, \mathbb{N}, \mathbb{Q}$ have infinite cardinalities.
- ▶ Every element of $\mathbb{Z}, \mathbb{N}, \mathbb{Q}$ can be mapped onto some value of $\mathbb{N}$, but *not* every element of $\mathbb{R}$ (e.g. π)
- ▶ An infinite set for which each element can be mapped onto a natural number is **countably infinite**. Otherwise, it is **uncountably infinite**.

Note that $\mathbb{Z}, \mathbb{N}, \mathbb{Q}$ all have the same cardinality (countable infinity), even though we can construct elements of $\mathbb{Z}, \mathbb{Q}$ which are not elements of $\mathbb{N}$. Then, we need another definition....

Closed vs. Open Sets

Not all infinite sets are created equally.

Think: are $\mathbb{N}$ and $x \in \mathbb{R} : \{0 \leq x \leq 1\}$ similar in structure?

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What about $\mathbb{N}$ and $x \in \mathbb{R} : \{0 < x < 1\}$?

We can instead define an **open set** as one where no element is a clear boundary (and a **closed set** as its alternative).

More Rigorously

A set is open iff around every element of that set, you can always construct a **neighborhood** that is entirely within that set.

That is, A is an open set if, for every $x \in A$, there exists some set B such that $x \in B$ and $B \subset A$.

Products of Sets

Oftentimes, we want to consider a set of all possible combinations of sets. We do so using the $\times$ operator:

$$\{1, 2\} \times \{A, B\} = \{(1, A), (1, B), (2, A), (2, B)\}$$

When the sets are identical, we can just use exponents as shorthand

Set Products & Examples

In formal theory, we might often consider a “state space.”

A canonical example (largely driven by Acemoglu & Robinson) considers the organization of the poor (P) and the elite (E) as either organized (h) or unorganized (ℓ). The state space follows:

$$\begin{aligned}\Omega &= \{\text{Poor organization}\} \times \{\text{Elite organization}\} \\ &= \{P^h, P^\ell\} \times \{E^h, E^\ell\} \\ &= \{(P^h, E^h), (P^h, E^\ell), (P^\ell, E^h), (P^\ell, E^\ell)\}\end{aligned}$$

What are we encoding?

Key Takeaways:

- ▶ Math is a language – precise, but not always consistent
- ▶ Focus on reading, not memorizing
- ▶ Let simplicity drive notation